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import pandas as pd
from sqlalchemy import create_engine

data = pd.read_csv('customer_shopping_behavior.csv')

# filling null values
data['Review Rating'] = data.groupby('Category')['
Review Rating'].transform(lambda x:x.fillna(x.median
()))
print(data.isnull().sum())

data.columns = data.columns.str.lower()
data.columns = data.columns.str.replace(' ','_')
data = data.rename(columns={'purchase_amount_(usd)':'
purchase_amount'})
print(data.columns)

# creating a column of different age group
labels = ['young adult','adult','middle-age','senior']
data['age_group'] = pd.qcut(data['age'],q=4, labels =
labels)
print(data[['age','age_group']].head(10))

#create column purchase_frequency_days
frequency_mapping = {
    'Fortnightly': 14,
    'Monthly': 30,
    'Weekly': 7,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}
data['purchase_frequency_days'] = data['
frequency_of_purchases'].map(frequency_mapping)
print(data[['purchase_frequency_days','
frequency_of_purchases']].head(10))

print(data[['discount_applied','promo_code_used']].head
(10))
print((data['discount_applied']==data['promo_code_used'
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]).all())

data= data.drop('promo_code_used',axis=1)

print(data.columns)

# mysql connection
username = 'root'
password = 'mahima20718'
host = '127.0.0.1'
port = '3306'
database = "customer_behavior"

engine = create_engine(f"mysql+pymysql://{username}:{
password}@{host}:{port}/{database}")

#write dataframe to mysql
table_name = "mytable"
data.to_sql(table_name, engine, if_exists="replace",
index=False)

#read back sample
pd.read_sql("select * from mytable LIMIT 5;", engine)
```